

1. Building BI Solutions

OLAP (Online Analytical Processing)

- ❖ Creating **multidimensional cubes** or **tabular models** that allow users to slice, dice, drill down, and pivot data quickly.
- ❖ Example: Using **SQL Server Analysis Services (SSAS)** or Power BI's tabular models for fast analytics.

ETL (Extract, Transform, Load)

- ❖ Designing and implementing workflows to move raw data from multiple sources (databases, APIs, flat files) into a structured data warehouse.
- ❖ Tools: **SSIS**, Azure Data Factory, Talend, Informatica, Power Query (M Language).

MicroStrategy (or similar BI platforms)

- ❖ Building enterprise BI dashboards, reports, and self-service analytics on platforms beyond Power BI.
- ❖ Skills overlap with Power BI but often involve additional features like automated distribution, enterprise security, and governed datasets.

2. Creating Visual Reports

- ✓ Designing interactive dashboards and visualizations in **Power BI** or other BI tools.
- ✓ Following **data visualization best practices**:
 - ❖ Choose the right chart for the right metric.
 - ❖ Maintain consistency in colors, labels, and formatting.
 - ❖ Optimize for performance and usability.
- ✓ Examples: KPI cards, trend lines, drill-through pages, slicers, heatmaps.

3. Documentation

- ✓ Preparing **technical documentation**:
 - ❖ ETL workflows
 - ❖ Data models & schema diagrams
 - ❖ DAX/M scripts & calculations
- ✓ Preparing **user documentation**:
 - ❖ Step-by-step usage guides for reports
 - ❖ Data dictionary explaining each metric
 - ❖ Access permissions and security notes

In a CV You could phrase it as:

"Designed and implemented BI solutions involving OLAP cubes, ETL pipelines, and MicroStrategy dashboards; developed interactive Power BI visual reports; maintained comprehensive technical and user documentation."

OLAP cubes (Online Analytical Processing cubes) are **multidimensional data structures** used in Business Intelligence to enable fast and flexible analysis of large amounts of data from multiple perspectives.

They are widely used in tools like **Power BI, SQL Server Analysis Services (SSAS), MicroStrategy, Cognos, and SAP BW** for advanced reporting and analytics.

1. What is an OLAP Cube?

A **data cube** that organizes data into **dimensions** (like Date, Product, Region) and **measures** (like Sales, Quantity, Profit).

Allows **slicing, dicing, drill-down**, and **pivoting** for exploring data interactively.

2. OLAP Cube Structure

1. Measures (Facts)

Quantitative values to analyze (e.g., Sales Amount, Revenue, Units Sold).

2. Dimensions

Categories for analysis (e.g., Time, Geography, Product, Customer).

3. Hierarchies

Dimension levels for drill-down (e.g., Year → Quarter → Month → Day).

4. Members

Individual items in a dimension (e.g., "India" in the Country dimension).

3. Types of OLAP

Type	Description	Example Use
MOLAP	Multidimensional OLAP — Data stored in a cube structure	Fast queries on summarized sales data
ROLAP	Relational OLAP — Data stored in relational tables, cube logic applied on the fly	Direct queries on SQL DB for detailed transactions
HOLAP	Hybrid OLAP — Combines MOLAP (aggregates) and ROLAP (detailed data)	Large datasets with summary + drill-through

4. Key OLAP Cube Operations

Operation	Description	Example
Slice	Filter cube on one dimension	Sales in 2024
Dice	Filter cube on multiple dimensions	Sales in 2024 for Electronics in India
Drill-Down	Go from summary to detail	Year → Month sales

Operation	Description	Example
Roll-Up	Aggregate details	Month → Year
Pivot	Rotate cube view	Switch rows/columns (Region ↔ Product)

5. Advantages

- ❖ Very **fast query performance** (pre-aggregated data).
- ❖ Flexible **multi-perspective analysis**.
- ❖ Handles **large datasets** efficiently.
- ❖ Ideal for **trend analysis, forecasting, and KPI monitoring**.

6. Example

Imagine a **Sales Cube**:

Dimensions: Time, Product, Region **Measures:** Sales Amount, Units Sold

You could quickly answer:

- ❖ "What were the sales of Laptops in India in Q1 2024?"
- ❖ "Which region had the highest profit last year?"
- ❖ "How do sales trends vary month-to-month for Mobile Phones?"

OLAP (Online Analytical Processing)

Definition: OLAP is a category of software tools that enables users to **analyze large volumes of data interactively** from multiple perspectives.

Key Concept:

- ❖ The data is stored in **multidimensional cubes** (or tabular models) rather than in flat relational tables.
- ❖ These cubes are designed to support **fast querying** and **complex aggregations**.

Multidimensional Cubes

- ❖ **Structure:** Data is organized into **dimensions** (e.g., Time, Geography, Product) and **measures** (e.g., Sales, Profit).
- ❖ **Purpose:** Enables slicing (filtering), dicing (changing dimensions), drilling down (from year → month → day), and pivoting (changing the orientation of the view).
- ❖ **Example:** A cube might store sales data by **Region, Product Category, and Year** so users can quickly view:
 - ❖ "Total Sales in North Region for Electronics in 2024"
 - ❖ "Quarterly Profit Trends for All Regions"

Tabular Models

- ❖ **Alternative to Multidimensional:** Uses relational tables with relationships (similar to a star schema).
- ❖ Stores data in **in-memory columnar storage** for speed.
- ❖ Often simpler and faster to develop, especially in tools like **Power BI** or **SSAS Tabular**.

How OLAP Helps in BI

- ❖ **Speed:** Pre-aggregated data allows instant queries.
- ❖ **Flexibility:** Users can change their view without new queries from IT.
- ❖ **Interactivity:** Supports drill-downs, roll-ups, and pivot operations directly in reports or dashboards.

Real-World Example: A **Power BI** report connected to an **SSAS Tabular Model** that allows a sales manager to:

- ❖ Filter by **Country → State → City**.
- ❖ Compare **Year-over-Year growth** instantly.

- ❖ Switch from viewing **Revenue** to **Profit Margin** with one click.

SSRS – SQL Server Reporting Services

Type: Microsoft server-based **reporting** platform.

Purpose: To create, manage, and deliver **paginated reports** from structured data sources.

Key Features

- ❖ **Report Design:** Create fixed-layout, printable reports (PDF, Excel, Word, HTML).
- ❖ **Data Sources:** SQL Server, Oracle, ODBC, OLE DB, XML, and more.
- ❖ **Deployment:** Host reports on a central SSRS server so users can access them via browser.
- ❖ **Delivery:** Scheduled email subscriptions, file shares, or on-demand viewing.
- ❖ **Customization:** Parameters, filters, drill-through reports, sub-reports.

Example Use: Generate monthly **sales invoices** for all customers, with exact formatting for printing or email distribution.

SSIS – SQL Server Integration Services

Type: Microsoft ETL (**Extract, Transform, Load**) platform. **Purpose:** To **move** and **transform** data between systems.

Key Features

- ❖ **Data Extraction:** Pull data from multiple sources (databases, flat files, Excel, APIs).
- ❖ **Data Transformation:** Clean, aggregate, merge, or reformat data.
- ❖ **Data Loading:** Push into a destination (SQL Server, data warehouse, cloud).
- ❖ **Automation:** Schedule ETL jobs via SQL Server Agent.
- ❖ **Components:** Control Flow (workflow), Data Flow (actual ETL), Parameters, Variables.

Example Use: Every night:

- ✓ Extract **sales transactions** from multiple store databases.
- ✓ Transform data into a standard format.
- ✓ Load into a central **data warehouse** for Power BI reporting.

SSRS vs SSIS – Quick Comparison

Feature	SSRS (Reporting)	SSIS (Integration)
Purpose	Generate and deliver reports	Move and transform data (ETL)
Output	Reports (PDF, Excel, HTML, Word)	Data pipelines
Focus	Presentation and visualization of data	Data extraction, transformation, and loading
Integration with BI	Can feed Power BI with paginated reports	Can prepare data for Power BI and ADF

4 How They Work Together

- ❖ SSIS prepares the data (extract from multiple sources, clean, load into SQL Server).
- ❖ SSRS consumes that prepared data to create professional, printable reports.

Relation to Modern Tools:

- ❖ SSIS overlaps with **Azure Data Factory** for ETL (ADF is cloud-based, SSIS is on-premises).
- ❖ SSRS overlaps with **Power BI paginated reports** (Power BI is more interactive, SSRS is better for fixed layouts).

1. Talend – Basics

Type: Open-source & commercial **ETL (Extract, Transform, Load)** tool. **Purpose:** Move and transform data between systems (databases, cloud, flat files, APIs).

Key Points:

- ❖ **Open-source roots** → Talend Open Studio (free) + enterprise editions.
- ❖ **Drag-and-drop interface** → design data workflows without heavy coding.
- ❖ **Connectivity** → supports databases, cloud storage, big data (Hadoop, Spark), REST/SOAP APIs.
- ❖ **Job Design:** Create ETL “jobs” using components like:
 - ❖ **tInput** → to read data
 - ❖ **tMap** → to transform/clean data
 - ❖ **tOutput** → to write data into the destination
- ❖ **Languages:** Generates **Java code** in the background.
- ❖ **Best For:** Small to mid-size ETL, cloud migrations, API integrations.

2. Informatica – Basics

Type: Commercial **ETL and Data Integration** platform (leader in enterprise BI/ETL).

Purpose: Large-scale, enterprise-grade data movement and transformation.

Key Points:

- ❖ **Enterprise focus** → used by banks, telecom, healthcare, manufacturing.
- ❖ **Components:**
 - ❖ **PowerCenter Designer** → build mapping between source & target.
 - ❖ **Workflow Manager** → schedule & control data jobs.
 - ❖ **Repository Manager** → manage metadata.
- ❖ **Connectivity** → Databases, flat files, ERPs, cloud storage, big data.
- ❖ **Advanced Features:** Data Quality, Data Governance, Metadata Management.

- ❖ **Best For:** Very large datasets, complex transformations, mission-critical ETL pipelines.

Talend vs Informatica – Quick Comparison

Feature	Talend (Open Studio)	Informatica PowerCenter
License	Open-source + Paid editions	Paid, enterprise-level
Ease of Use	Easy drag-and-drop UI	Easy but more structured
Best For	Small to mid-scale projects	Large enterprise integrations
Performance	Good, Java-based	Very high, optimized engine
Cloud/Big Data	Strong integration (Spark, AWS)	Strong but more enterprise-focused

Real-World ETL Example in Both Tools:

- ❖ **Extract:** Pull data from MySQL + Salesforce
- ❖ **Transform:** Clean missing values, merge datasets, calculate KPIs
- ❖ **Load:** Store into Azure SQL Data Warehouse for Power BI dashboards.

Installing Talend Open Studio (the free version) on Windows

Check System Requirements

- ❖ **OS:** Windows 10/11 (64-bit), macOS, or Linux
- ❖ **RAM:** 4 GB minimum (8 GB recommended)
- ❖ **Java:** JDK 8 or higher (most stable with Talend)
- ❖ **Disk Space:** At least 3 GB free

Install Java (Required)

Talend Open Studio runs on Java, so you must install it first.

Steps:

- ❖ Go to: Java SE Downloads
- ❖ Download **Java JDK 8** (64-bit).
- ❖ Install it on your system.
- ❖ Set the **JAVA_HOME** environment variable:
 - ❖ Search **Edit the system environment variables** → **Environment Variables**
 - ❖ Under **System Variables**, click **New** → Name: JAVA_HOME, Value: your JDK folder path (e.g., C:\Program Files\Java\jdk1.8.0_371).
 - ❖ Add %JAVA_HOME%\bin to **Path**.

Download Talend Open Studio

- ❖ Go to **Talend official site**: Talend Downloads
- ❖ Choose **Talend Open Studio for Data Integration** (free).
- ❖ Register with your email (required for download).
- ❖ Download the .zip file.

Extract Talend

Right-click the downloaded .zip → **Extract All...**

Choose a location like C:\Talend (avoid spaces in path).

Launch Talend

- ✓ Open the extracted folder.

- ✓ Run:
 - **Windows:** TOS_DI-win-x86_64.exe
 - **Mac:** TOS_DI-macosx-cocoa.app
- ✓ Select a **workspace** folder (where your projects will be saved).
- ✓ Wait for Talend to load (first launch may take 1–2 minutes).

6First-Time Setup

- ❖ Accept the license agreement.
- ❖ Create a new **local project** (you can skip remote connection).
- ❖ Click **Finish** and start building ETL jobs.

Quick Tip: If Talend doesn't start, check that:

- ❖ You're using a compatible Java version (JDK 8 is safest).
- ❖ JAVA_HOME is set correctly.
- ❖ You have extracted the zip fully (not running from inside zip).

What is Azure Data Factory (ADF)?

- ❖ **Type:** Cloud-based ETL (Extract, Transform, Load) & ELT (Extract, Load, Transform) service.
- ❖ **Purpose:** To **move**, **transform**, and **integrate** data from multiple sources into storage or analytics systems.
- ❖ **Key Benefit:** No need to manage servers — it's fully managed in Azure.

Core Use Cases

- ❖ Data migration from **on-premises to cloud**.
- ❖ Integrating **multiple data sources** (databases, APIs, files, cloud storage).

- ❖ **Big data processing** (with Azure Databricks, HDInsight).
- ❖ Automating **data pipelines** for reporting & analytics (e.g., Power BI dashboards).

Main Components

Component	Purpose
Pipelines	A logical group of activities that perform a task (like a workflow).
Activities	Steps inside a pipeline (e.g., copy data, run stored procedure, transform data).
Datasets	Data structures that point to the data you want to use (e.g., a table, a file).
Linked Services	Connection info to a data source (e.g., SQL Server, Blob Storage).
Triggers	Define when a pipeline should run (schedule, event-based).
Integration Runtime (IR)	The compute infrastructure used to run activities.

How It Works

- ❖ **Connect** to data sources (on-premises, cloud, SaaS apps).
- ❖ **Ingest** data into staging or storage (e.g., Azure Data Lake, Blob).
- ❖ **Transform** data using:
 - Mapping Data Flows (drag-and-drop transformations)
 - Azure Databricks (PySpark, SQL, ML)
 - Stored Procedures or external compute
- ❖ **Load** into target system (SQL DB, Synapse, Power BI, etc.).
- ❖ **Schedule** using triggers for automation.

Supported Data Sources

- ❖ **On-premises:** SQL Server, Oracle, MySQL, SAP
- ❖ **Cloud:** Azure Blob, Data Lake, Amazon S3, Google Cloud Storage
- ❖ **SaaS:** Salesforce, Dynamics 365, Google Analytics, REST APIs

6 Example Workflow

Imagine you want to update a **Power BI sales dashboard** every morning:

- ❖ **Trigger:** Runs at 6:00 AM daily.
- ❖ **Copy Activity:** Pulls sales data from an on-premises SQL Server.
- ❖ **Mapping Data Flow:** Cleans and aggregates sales data.
- ❖ **Load:** Saves results in Azure SQL Database.
- ❖ **Power BI Refresh:** Dashboard updates with new data.

Key Advantages

- ❖ **Scalable:** Handles small & huge datasets.
- ❖ **Serverless:** No hardware to maintain.
- ❖ **Cost-Effective:** Pay only for what you use.
- ❖ **Hybrid Capability:** Works with both cloud and on-premises sources.

Working with Azure Data Factory (ADF) — from account setup to creating your first pipeline connection.

Prerequisites

- ❖ **Azure Subscription** (Free or Paid)

If you don't have one, sign up here: Azure Free Account

- ❖ Basic familiarity with cloud portals and data sources you'll connect to.
- ❖ Access credentials for your **source** (e.g., SQL Server, Blob Storage) and **destination**.

Log in to Azure Portal

Go to **portal.azure.com**

Sign in with your Azure account.

Create an Azure Data Factory

5. In the search bar, type **Data Factory** and click **Create**.

6. Fill in details:

Subscription: Choose your Azure subscription.

Resource Group: Select an existing one or create a new group.

Region: Choose the closest data center.

Name: Unique ADF name (e.g., myadf-demo).

Version: Always choose **V2**.

7. Click **Review + Create** → **Create**.

Open Azure Data Factory Studio

Once deployed, click **Go to Resource**.

Click **Open Azure Data Factory Studio** — this opens the web-based interface for building pipelines.

Create a Linked Service (Connection)

Linked Services are **connection definitions** for your data sources and destinations.

Example: Connect to Azure Blob Storage

In ADF Studio, go to **Manage** (left menu).

Under **Linked services**, click **New**.

Choose **Azure Blob Storage**.

Enter:

- ❖ **Name:** LS_BlobStorage
- ❖ **Authentication:** Account key, SAS token, or Azure AD.
- ❖ **Storage account name:** Select your Blob Storage.

8. Click **Test Connection** → **Create**.

Create a Dataset

Datasets define the **specific data structure** you want to work with inside a Linked Service.

Example: CSV file in Blob Storage

In **Author** (pencil icon), click + → **Dataset**.

Choose **Azure Blob Storage** → Format: **DelimitedText (CSV)**.

Link it to the Linked Service (LS_BlobStorage).

Select the container and file path.

Build a Simple Pipeline

- ❖ In **Author**, click + → **Pipeline**.
- ❖ Search for **Copy Data** activity and drag it into the canvas.
- ❖ Set:

- **Source:** The Dataset from Blob.
 - **Sink:** Another Dataset (e.g., Azure SQL Database).
- ❖ Validate and **Debug** to test.

Trigger & Schedule

Click **Trigger** → **Trigger Now** to run instantly.

Or use **Triggers** for scheduled or event-based runs.

What is MicroStrategy?

- ❖ A **BI tool** that connects to multiple data sources (databases, APIs, cloud, Excel, big data platforms).
- ❖ Allows **data exploration, dashboard creation, OLAP analysis, predictive analytics,** and **mobile BI.**
- ❖ Known for **scalability** and handling very large enterprise datasets.

2. Key Features

Feature	Description
Interactive Dashboards	Build visually rich dashboards with charts, tables, KPIs, and filters.
OLAP Analysis	Slice, dice, drill-down, and pivot on multidimensional datasets.
Mobile BI	Native mobile apps for iOS/Android for interactive reports.
Embedded Analytics	Integrate analytics into custom apps or websites.
AI & Predictive Analytics	Integrates with R, Python, and ML models.
HyperIntelligence	Shows instant insights in any application via contextual pop-ups.
Cloud & On-Premises	Available as both SaaS and on-prem installations.

3. MicroStrategy Architecture

- ✓ **Metadata Layer** – Stores definitions of reports, dashboards, metrics, and datasets.
- ✓ **Intelligence Server** – Executes queries, applies business logic, and delivers results.
- ✓ **Data Sources** – Connects to RDBMS (Oracle, SQL Server, MySQL), big data (Hadoop), cloud (AWS, Azure, GCP), APIs.
- ✓ **Clients** – Web, Desktop, and Mobile interfaces for end-users.

4. OLAP in MicroStrategy

- ✓ Uses **Intelligent Cubes** (like OLAP cubes) for **fast query response**.
- ✓ Supports:
 - ❖ **Slicing/Dicing** (filtering)
 - ❖ **Drill-Down & Roll-Up** (hierarchies like Year → Month → Day)
 - ❖ **Pivoting** (changing the view)
- ✓ Ideal for **financial reports, sales dashboards, and KPI monitoring**.

5. Advantages

- ❖ Enterprise-grade **scalability**.
- ❖ Works with **massive datasets**.
- ❖ Highly **customizable dashboards**.
- ❖ **Real-time analytics** capability.
- ❖ Strong **security and governance**.

6. Example Use Cases

- ❖ **Retail** – Sales performance by region, product, and store.
- ❖ **Banking** – Loan portfolio risk analysis.

- ❖ **Healthcare** – Patient outcomes tracking.
- ❖ **Telecom** – Network performance monitoring.

You can install MicroStrategy in two main ways:

- ❖ **Cloud version** (quickest — no local installation)
- ❖ **Desktop version** (local installation, good for personal BI work)

Below are the **step-by-step installation options**.

Option 1: MicroStrategy Cloud (No Installation)

If you want to start quickly without downloading:

- ❖ Go to **<https://www.microstrategy.com/en/get-started>**
- ❖ Click "**Try MicroStrategy for Free**".
- ❖ Sign up with your **work or personal email**.
- ❖ You'll get **instant access** to MicroStrategy Workstation & Web in the cloud.
- ❖ Start building dashboards directly in the browser.

Best for: Beginners or testing without setup.

Option 2: Install MicroStrategy Desktop (Free Personal Edition)

Steps:

Visit <https://www.microstrategy.com/en/download>.

Under "**MicroStrategy Desktop**", click **Download Now**.

Choose your OS:

Windows (.exe file)

Mac (.dmg file)

Install it like any other app:

Windows: Double-click .exe → Follow setup wizard.

Mac: Drag the app into **Applications**.

Launch **MicroStrategy Desktop**.

Sign in with your MicroStrategy account (create one if new).

Connect to your **data source** and start building dashboards.

Option 3: Full Enterprise / Server Installation (For IT Admins)

If you want OLAP cubes, enterprise security, and multi-user dashboards:

Download **MicroStrategy Server** from:

<https://www.microstrategy.com/en/download>

Install **MicroStrategy Intelligence Server** and **Web** components.

Set up **metadata & warehouse databases**.

Configure **Intelligent Cubes** for OLAP.

Connect clients (Web, Desktop, Mobile).

Note: This requires **Windows Server** (or Linux), SQL/Oracle DB, and IT admin rights.

Minimum System Requirements for Desktop

- ❖ **OS:** Windows 10/11 or macOS 10.15+
- ❖ **RAM:** 8 GB (16 GB recommended)
- ❖ **Disk Space:** 2 GB free
- ❖ **Processor:** 64-bit Intel or AMD CPU
- ❖ **Internet:** Needed for cloud sign-in and updates